MILITARY SPECIFICATION
FIRING MECHANISM, CARTRIDGE ACTUATED: 29378-1 ASSEMBLY

1. SCOPE:

1.1 This specification covers Firing Mechanism 29378-1. The assembly will be used to activate the canopy counter balance jettison and cinch down cylinder assembly in the TF-102 Aircraft.

2. APPLICABLE DOCUMENTS:

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for.
MIL-L-4343	- Lubricating Grease, Pneumatic System

STANDARDS
Military

MIL-STD-105	- Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

C9224217	- Container, Ammunition, Fiber, for Firing Mechanism Assembly, P/N 29378-1.
C9224720	- Box, Packing, Ammuniton, for Firing Mechanism Assembly, P/N 29378-1.
IEL-29378-1	- Inspection Equipment List for Firing Mechanism, Cartridge Actuated 29378-1 Assembly.

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2.1 (Cont'd)

Talley Industries (Code Ident No. 12116)

29378-1 - Firing Mechanism, Cartridge Actuated, Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply:

Interstate Commerce Commission - 49 CFR 71-78

The Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and other Dangerous Articles. (The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. Orders for above publication should cite "49 CFR 71-78").

3. REQUIREMENTS:

3.1 General - Material, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Torque - Torque requirements shall be as specified in the applicable drawings.

3.3 Residual magnetism - The firing mechanism shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional:

3.4.1 The firing mechanism shall not fail to fire.

3.4.2 The firing mechanism shall function without mechanical failure (See 6.2).

3.4.3 Initiation - The firing mechanism shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size 4, MS28741, at 70° ± 5°F. (Max 500 psig).

3.4.4 Ballistics - The firing mechanism loaded with an accepted cartridge and temperature conditioned at -65°F, 70°F and 200°F shall be fired into a 5.60 cubic inch volume for pressure-time performance and data recorded for information.

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TABLE I

Parameter	Unit of Measurement	-65° to -70°F Min	-65° to -70°F Max	65° to 75°F Min	65° to 75°F Max	200° to 205°F Min	200° to 205°F Max
Ignition Delay	SEC	*	*	*	*	*	*
Time to Peak Pressure	SEC	*	*	*	*	*	*
Peak Pressure	PSIG	*	*	*	*	*	*

* For Information Only.

3.5 Workmanship -

3.5.1 Metal defects - All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defects.

3.5.2 Burr - Parts shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter - Parts or the assembly shall not contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions - Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions

4.2.1 Lot

4.2.1.1 Submission of product - The product shall be submitted in accordance with MIL-STD-105.

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4.2.1.2 Size of lot - A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of accepted cartridges.

4.2.2 Examination - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming material shall be rejected. Four power magnification may be used in performing visual examination for inspection of such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3
1801 -10000	200	2	3	3	4

4.2.2.1 Cartridge Assembly, P/N 29309-0, covering a detail of P/N 29378-1

	Defect	Method of Inspection
CRITICAL		
1.	Mixed cartridges	Visual
2.	Metal defective	Visual
3.	Sealing disc missing or damaged	Visual
4.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Foreign matter	Visual
MINOR		
201.	Case scratched, wrinkled, bulged or dented	Visual
202.	Protective finish, missing or inadequate	Visual

4.2.2.2 Assembly, P/N 29378-1, less P/N AN565D8H2

CRITICAL		
1.	Firing pin missing	Visual
2.	Shear pin missing or not in groove	Visual

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4.2.2.2 Assembly, P/N 29378-1, less P/N AN565D8H2 - Cont'd

Defect	Method of Inspection
MAJOR	
None defined	
MINOR	
None defined	

4.2.2.3 Assembly P/N, 29378-1

CRITICAL		
1.	Improper assembly	Visual
2.	Any parts missing	Visual
3.	Labels missing, incorrect or illegible	Visual
MAJOR		
101.	Proper alignment	Visual
102.	Set screw missing, loose, above flush, or not covered with sealing compound	Visual
MINOR		
201.	Location of labels incorrect	Visual
202.	Proective finish missing or inadequate	Visual
203.	Shipping cap missing	Visual

4.2.2.4 Container, Ammunition, Fiber, Drawing C9224217

CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Gap between body and cover, max.	Manual
202.	Contents loose	Manual
203.	Metal ends loose or distorted	Manual
204.	Cut or damaged through all impregnated layers	Visual

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4.2.2.4 Container, Ammunition, Fiber, Drawing C9224217 - Cont'd

Defect	Method of Inspection
205. Marking, missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.5 Box, Packing, Ammunition, Drawing C9224720

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Hardware missing, loose, broken, or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Marking missing, incorrect or illegible	Visual
206. Number of units incorrect	Visual
207. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection - The following tests shall be performed in accordance with 4.3.

4.2.3.1 Assembly torque - Threaded joints of each assembly shall be subjected to the torque application. Failure of any assembly to comply with the requirement of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism - Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.3 shall be cause for rejection of the assembly.

4.2.3.3 Functional tests - The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics - Twenty-one units shall be tested as specified in 4.3.3 to determine compliance with the requirements of 3.4.1 to 3.4.4. The units shall be temperature conditioned as provided in Table III and 4.3.3.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

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TABLE III

Temperature	Sample Size
-65° to -70°F	7
65° to 75°F	7
200° to 205°F	7

NOTE: All variable data gained at -65°F and 200°F will be recorded and reported for information only.

4.2.4 Inspection equipment - The examination and tests shall be made using equipment approved by the Government.

4.3 Test methods and procedures

4.3.1 Assembly torque - A torque wrench, set at the torque requirements specified in 3.2 shall be used in conjunction with an adapter to determine the adequacy of the joints. The firing mechanism shall be secured in a holding device and the assembly torque applied to the head.

4.3.2 Residual magnetism - The assembly shall be tested for excessive residual magnetism, in an area free of local magnetic effects, by placing the assembly six inches from and at the same height as the compass in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly past the compass. This procedure shall be repeated for each 90 degree rotation of the firing mechanism.

4.3.3 Functional tests

4.3.3.1 Temperature conditioning - Prior to conducting the ballistic test specified in 4.3.3.2, the assembly samples shall be subjected to the temperature condition specified in 3.4.4. A simulated assembly loaded with a dummy cartridge containing a temperature sensitive element shall be placed in the temperature conditioning chamber at the same time as the assembly samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.3.3.1.1 Upon completion of the required temperature conditioning, each assembly sample shall be ballistically tested within 5 minutes after removal from the conditioning chamber. Samples shall be reconditioned if this time is exceeded.

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4.3.3.2 Method of testing - The firing mechanism shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4, MS 28741 to determine compliance with 3.4. The firing mechanism shall be rigidly supported in a test fixture and fired into a 5.60 cubic inch volume for pressure time performance.

5. PREPARATION FOR DELIVERY:

5.1 Packing and marking - Level A - Packing and marking shall be in accordance with drawings C9224217 and C9224720.

6. NOTES:

6.1 Ordering data - Procurement documents should specify the title number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

Preparing activity:

Army - MU - FA

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